

Cisco Configuration Guide

Router Configuration

enable

configure terminal

hostname R1

interface gigabitethernet 0/0

no shutdown

exit

interface gigabitethernet 0/0.10

encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0

exit

interface gigabitethernet 0/0.20

encapsulation dot1Q 20

ip address 192.168.20.1 255.255.255.0

exit

interface gigabitethernet 0/0.30

encapsulation dot1Q 30

ip address 192.168.30.1 255.255.255.0

exit

Switch Configuration

enable

configure terminal

hostname SW1

vlan 10

name ADMIN

exit

vlan 20

name IT

exit

vlan 30

name GUEST

exit

Admin Port

interface fastethernet 0/1

switchport mode access

switchport access vlan 10

exit

IT Port

interface fastethernet 0/2

switchport mode access

switchport access vlan 20

exit

Guest Port

interface fastethernet 0/3

switchport mode access

switchport access vlan 30

exit

Trunk

interface gigabitethernet 0/1

switchport mode trunk

exit

Extended ACL

```
ip access-list extended GUEST-RESTRICTION
deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
permit ip 192.168.30.0 0.0.0.255 any
exit
```

Apply the ACL:

```
interface gigabitethernet 0/0.30
ip access-group GUEST-RESTRICTION in
exit
```

Verification

```
show vlan brief
show interfaces trunk
show ip interface brief
show ip route
show access-lists
```